PRACTICE SHEET 1

1. The value of $\sqrt{\dfrac{(0.03)^2+(0.21)^2+(0.065)^2}{(0.003)^2+(0.021)^2+(0.0065)^2}}$

(1) 0.1 (2) 10
(3) 10^2 (4) 10^3

2. The sum of $\sqrt{0.01}+\sqrt{0.81}+\sqrt{1.21}+\sqrt{0.0009}$ is

(1) 2.1 (2) 2.13
(3) 2.03 (4) 2.11

3. The value of $\sqrt{\dfrac{(6.1)^2+(61.1)^2+(611.1)^2}{(0.61)^2+(6.11)^2+(61.11)^2}}$

(1) 0.1 (2) 1.1
(3) 10 (4) 100

4. The value of $\sqrt{5+\sqrt{11+\sqrt{19+\sqrt{29+\sqrt{49}}}}}$ is

(1) 3 (2) 9
(3) 7 (4) 5

5. $\sqrt{\dfrac{0.081\times0.484}{0.0064\times6.25}}$ is equal to

(1) 9 (2) 0.9
(3) 99 (4) 0.99

6. The simplified value of $\sqrt{900}+\sqrt{0.09}-\sqrt{0.000009}$ is

(1) 30.27 (2) 30.297
(3) 30.097 (4) 30.197

7. Which of the following is not a perfect square

(1) 678976 (2) 157609
(3) 536582 (4) 419904

8. The smallest 4-digit number, which is a perfect square, is

(1) 1009 (2) 1016
(3) 1024 (4) 1025

9. The smallest number added to 680621 to make the sum a perfect square is :

(1) 4 (2) 5
(3) 6 (4) 8

10. The smallest positive integer, when multiplied by 392, the product is a perfect square, is

(a) 6 (b) 5
(c) 3 (d) 2

11. Which smallest number must be added to 2203 so that we get a perfect square?

(a) 1 (b) 3
(c) 6 (d) 8

12. The number of perfect square numbers between 50 and 1000 is

(a) 21 (b) 22
(c) 23 (d) 24

13. The smallest number which should be added to the number 8958 so that the result is a perfect square is

(a) 69 (b) 67
(c) 77 (d) 79

14. The largest number of five digits, which is a perfect square is

(a) 99999 (b) 99976
(c) 99856 (d) 99764

15. How many perfect squares lie between 120 and 300 ?

(a) 5 (b) 6
(c) 7 (d) 8

16. The smallest number that must be subtracted from 1000 to make the resulting number a perfect square is

(a) 37 (b) 38
(c) 39 (d) 40

17. $\sqrt{42+\sqrt{42+\sqrt{42+\ldots\ldots\ldots}}}=$

(a) 7 (b) 6
(c) 8 (d) 9

18. $\sqrt{56+\sqrt{56+\sqrt{56+\ldots\ldots\ldots}}}=$

(a) 7 (b) 6
(c) 8 (d) 9

19. $\sqrt{9\sqrt{9\sqrt{9\sqrt{9\ldots\ldots}}}}=$

(a) 7 (b) 6
(c) 8 (d) 9

20. $\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2+\ldots\ldots}}}}=$

(a) 0 (b) 1

(c) 2 (d) 3

21. What number must be added to the expression $16a^2 - 12a$ to make it a perfect square?
(1) 9/4 (2) 11/2
(3) 13/2 (4) 16

22. What value should come in place of x in $\frac{x^{\frac{9}{4}}}{324} = \frac{x^{\frac{1}{4}}}{9}$
(a) 27 (b) 36^2
(c) 6 (d) 36

23. One-third of the square root of which number is 0.001?
(1) 0.0009 (2) 0.000001
(3) 0.00009 (4) None of the above

24. $\sqrt{7+\sqrt{48}} =$
(a) $4 + \sqrt{2}$ (b) $2 + \sqrt{3}$
(c) $3 - \sqrt{2}$ (d) $4 - \sqrt{2}$

25. If the sum of the squares of three consecutive natural numbers is 110, then the smallest of these natural numbers is:
(1) 8 (2) 6
(3) 7 (4) 5

26. Which of the following is true?
(1) $\sqrt{5} + \sqrt{3} > \sqrt{6} + \sqrt{2}$
(2) $\sqrt{5} + \sqrt{3} < \sqrt{6} + \sqrt{2}$
(3) $\sqrt{5} + \sqrt{3} = \sqrt{6} + \sqrt{2}$
(4) $(\sqrt{5} + \sqrt{3})(\sqrt{6} + \sqrt{2}) = 1$

27. If 5416 * 6 is a perfect square, then the digit at * is :
(1) 9 (2) 4
(3) 6 (4) 5

28. $4(10 + 15 \div 5 \times 4 - 2 \times 2) =$
(a) 56 (b) 24
(c) 48 (d) 72

29. The sum of squares of three positive integers is 323. If the sum of squares of two numbers is twice the third, their product is
(1) 255 (2) 260
(3) 265 (4) 270

30. The difference between two numbers is 9 and the difference between their squares is 207. The numbers are:
(1) 17 and 8 (2) 16 and 7
(3) 15 and 6 (4) 23 and 14

ANSWER KEY																			
1.	(b)	2.	(b)	3.	(c)	4.	(a)	5.	(d)	6.	(b)	7.	(c)	8.	(c)	9.	(a)	10.	(d)
11.	(c)	12.	(d)	13.	(b)	14.	(c)	15.	(c)	16.	(c)	17.	(a)	18.	(c)	19.	(d)	20.	(c)
21.	(a)	22.	(c)	23.	(d)	24.	(b)	25.	(d)	26.	(a)	27.	(a)	28.	(d)	29.	(a)	30.	(b)

PRACTICE SHEET 2

1. If $x = 1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{2}}}}$ then the value of $2x+\frac{7}{4}$ is
(1) 3 (2) 4
(3) 5 (4) 6

2. The simplification of $\cfrac{5}{3+\cfrac{3}{1-\cfrac{2}{3}}}$ gives
(a) 5 (b) 5/3
(c) 5/12 (d) 3/5

3. If $2 = x + \cfrac{1}{1+\cfrac{1}{3+\cfrac{1}{4}}}$, then the value of x is
(a) 18/17 (b) 21/17
(c) 13/17 (d) 12/17

4. Simplify $1+\cfrac{4}{2+\cfrac{3}{5-\cfrac{1}{2}}}-\frac{1}{2}(10\div 2)$

(a) 1 (b) 0
(c) - 15/2 (d) - 1/2

5. The value of $1-\cfrac{a}{1-\cfrac{1}{1+\cfrac{a}{1-a}}}$ is
(a) a (b) 1 - a
(c) 1 (d) 0

6. Simplify:
$8\frac{1}{2}-\left[3\frac{1}{4}\div\left\{1\frac{1}{4}-\frac{1}{2}\left(1\frac{1}{2}-\frac{1}{3}-\frac{1}{6}\right)\right\}\right]$
(a) $4\frac{1}{2}$ (b) $4\frac{1}{6}$
(c) $9\frac{1}{2}$ (d) $\frac{2}{9}$

7. The value of 0.008 × 0.01 × 0.072 ÷ (0.12 × 0.0004) is:
(1) 1.2 (2) 0.12
(3) 0.012 (4) 1.02

8. The value of $\frac{2}{3}\times\frac{3}{\frac{5}{6}\div\frac{2}{3}\text{of}1\frac{1}{4}}$ is
(a) 2 (b) 1
(c) 1/2 (d) 2/3

9. $9-1\frac{2}{9}of\,3\frac{3}{11}\div5\frac{1}{7}of\,\frac{7}{9}$ is equal to
(a) 8 (b) 9
(c) 7 (d) $\frac{3}{4}$

10. The value of 1 ÷ [1 + 1 ÷ {1 + 1 ÷ (1 + 1 ÷ 2)}] is
(1) 1 (2) 5/8
(3) 2 (4) 1/2

11. $\frac{1}{2}+\frac{1}{6}+\frac{1}{12}+\frac{1}{20}+\frac{1}{30}+\frac{1}{42}=$
(a) 5/7 (b) 6/7
(c) 2/7 (d) 1

12. $\frac{1}{12}+\frac{1}{20}+\frac{1}{30}+\frac{1}{42}+\frac{1}{56}+\frac{1}{72}=$
(a) 2/9 (b) 4/9
(c) 3/4 (d) 5/9

13. $\frac{1}{1.3}+\frac{1}{3.5}+\frac{1}{5.7}+\frac{1}{7.9}+\frac{1}{9.11}=$
(a) 5/11 (b) 10/11
(c) 6/11 (d) 3/11

14. $\frac{1}{15}+\frac{1}{35}+\frac{1}{63}+\frac{1}{99}+\frac{1}{143}=$
(a) 6/39 (b) 10/39
(c) 5/39 (d) 4/39

15. $\frac{1}{6}+\frac{1}{12}+\frac{1}{20}+\frac{1}{30}+\frac{1}{42}+\frac{1}{56}=$
(a) 3/8 (b) 5/8
(c) 7/8 (d) 1/8

16. $(16)^{0.16}\times(16)^{0.04}\times(2)^{0.2}$ is equal to :
(1) 1 (2) 2
(3) 4 (4) 16

17. The square root of 14 + 6√5 is
(1) 2 + √5 (2) 3 + √5
(3) 5 + √3 (4) 3 + 2√5

18. $\left(\sqrt{2}+\sqrt{7-2\sqrt{10}}\right)$ is equal to
(a) √2 (b) √7
(c) √5 (d) 2√5

19. Which of the following is largest $\sqrt[3]{4}$, $\sqrt[4]{6}$, $\sqrt[6]{15}$, $\sqrt[12]{245}$
(a) $\sqrt[3]{4}$ (b) $\sqrt[4]{6}$
(c) $\sqrt[6]{15}$ (d) $\sqrt[12]{245}$

20. The value of $\frac{1}{\sqrt{2}+1}+\frac{1}{\sqrt{3}+\sqrt{2}}+\frac{1}{\sqrt{4}+\sqrt{3}}$
$\frac{1}{\sqrt{5}+\sqrt{4}}+\frac{1}{\sqrt{6}+\sqrt{5}}+\frac{1}{\sqrt{7}+\sqrt{6}}+\frac{1}{\sqrt{8}+\sqrt{7}}+\frac{1}{\sqrt{9}+\sqrt{8}}$
(a) 0 (b) 1
(c) 2 (d) 3

21. What is the square root of $9+2\sqrt{14}$?
(1) $1+2\sqrt{2}$ (2) $\sqrt{3}+\sqrt{6}$
(3) $\sqrt{2}+\sqrt{7}$ (4) $\sqrt{2}+\sqrt{5}$

22. The sum of the square of a number and the square of the reciprocal of the number is thrice the difference of the square of the number and the square of the reciprocal of the number. What is the number?
(1) 1 (2) $(2)^{1/4}$
(3) $(3)^{1/3}$ (4) $(4)^{1/4}$

23. What is/are the real value(s) of $(256)^{0.16}\times(16)^{0.18}$?
(1) Only – 4 (2) Only 4
(3) 4, – 4 (4) 2, – 2

24. Assertion (A): $\sqrt{\frac{5041}{6889}}$ is rational

Reason (R): The square root of a rational number is always rational.
(1) A and R are correct and R is correct explanation of A
(2) A and R are correct but R is not correct explanation of A
(3) A is correct but R is wrong
(4) A is wrong but R is correct

25. What is the square of $(2 + \sqrt{2})$?
(1) A rational number
(2) An irrational number
(3) A natural number
(4) A whole number

26. If $(ab^{-1})^{2x-1} = (ba^{-1})^{x-2}$
(1) 1 (2) 2
(3) 3 (4) 4

27. What is the value of $\sqrt{7.84} + \sqrt{0.0784} + \sqrt{0.000784} + \sqrt{0.00000784}$?
(1) 3.08 (2) 3.108
(3) 3.1008 (d) 3.1108

28. What is the value of $\frac{1}{1+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+........+\frac{1}{\sqrt{15}+\sqrt{16}}$
(1) 0 (2) 1
(3) 2 (4) 3

29. If $27\times(81)^{2n+3} - 3^m = 0$, then what is m equal to?
(1) $2n + 5$ (2) $5n + 6$
(3) $8n + 3$ (4) $8n + 15$

30. The number $\sqrt{0.0001}$ is
(a) a rational number less than 0.01
(b) a rational number
(c) an irrational number
(d)neither a rational number nor an irrational number

ANSWER KEY																			
1.	(3)	2.	(3)	3.	(2)	4.	(2)	5.	(4)	6.	(2)	7.	(2)	8.	(1)	9.	(1)	10.	(2)
11.	(2)	12.	(1)	13.	(1)	14.	(2)	15.	(1)	16.	(2)	17.	(2)	18.	(3)	19.	(1)	20.	(3)
21.	(3)	22.	(2)	23.	(2)	24.	(3)	25.	(2)	26.	(1)	27.	(4)	28.	(4)	29.	(4)	30.	(2)

CDS PYQ

1. What is the value of $\sqrt{6+\sqrt{6+\sqrt{6+..........}}}$
[CDS 2013(I)]
(a) 2 (b) 3
(c) 3.5 (d) 4

2. If $16\times8^{n+2} = 2^m$, then m is equal to
[CDS 2013(I)]
(a) $n + 8$ (b) $2n + 10$
(c) $3n + 2$ (d) $3n + 10$

3. The expression $\left[\left(\sqrt{2}\right)^{\sqrt{2}}\right]^{\sqrt{2}}$ gives
[CDS 2013(I)]
(a) a natural number
(b) an integer and not a natural number
(c) a rational number but not an integer
(d) a real number but not a rational number

4. If $x^2 = 6+\sqrt{6+\sqrt{6+\sqrt{6+..........}}}$, then what is one of the values of x equal to
[CDS 2013(II)]
(a) 6 (b) 5
(c) 4 (d) 3

5. Which is the smallest number among the following?
[CDS 2013(II)]
(a) $\left[\left(5^{-2}\right)^{-2}\right]^{-2}$ (b) $\left[\left(5^{-2}\right)^{2}\right]^{2}$
(c) $\left[\left(2^{-5}\right)^{-2}\right]^{-2}$ (d) $\left[\left(2^{-5}\right)^{2}\right]^{-2}$

6. Consider the following in respect of the numbers $\sqrt{2}$, $\sqrt[3]{3}$, $\sqrt[6]{6}$
1. $\sqrt[6]{6}$ is the greatest number

2. $\sqrt{2}$ is the smallest number

Which one of the above statements is/are correct ?

[CDS 2014(I)]

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

7. What is the value of $\frac{5+\sqrt{10}}{5\sqrt{5}-2\sqrt{20}-\sqrt{32}+\sqrt{50}}$ equal to

[CDS 2015(I)]

(a) 5 (b) $5\sqrt{2}$
(c) $5\sqrt{5}$ (d) $\sqrt{5}$

8. If x = $\sqrt{3}+\sqrt{2}$, then the value of $x^3+x+\frac{1}{x}+\frac{1}{x^3}$ is

[CDS 2015(II)]

(a) $10\sqrt{3}$ (b) $20\sqrt{3}$
(c) $10\sqrt{2}$ (d) $20\sqrt{2}$

9. Which one of the following is correct?

[CDS 2015(II)]

(a) $\sqrt{2}<\sqrt[4]{6}<\sqrt[3]{4}$ (b) $\sqrt{2}>\sqrt[4]{6}>\sqrt[3]{4}$
(c) $\sqrt[4]{6}<\sqrt{2}<\sqrt[3]{4}$ (d) $\sqrt[4]{6}>\sqrt{2}>\sqrt[3]{4}$

10. The sum of first 47 terms of series $\frac{1}{4}+\frac{1}{5}-\frac{1}{6}-\frac{1}{4}-\frac{1}{5}+\frac{1}{6}+\frac{1}{4}+\frac{1}{5}-\frac{1}{6}$....... is

[CDS 2015(II)]

(a) 0 (b) $-\frac{1}{6}$
(c) $\frac{1}{6}$ (b) $\frac{9}{20}$

11. The value of $\frac{1}{1\times4}+\frac{1}{4\times7}+\frac{1}{7\times10}+.......+\frac{1}{16\times19}$:

[CDS 2015 (II)]

(a) $\frac{5}{19}$ (b) $\frac{6}{19}$
(c) $\frac{8}{19}$ (d) $\frac{9}{19}$

12. What is $\frac{6^2+7^2+8^2+9^2+10^2}{\sqrt{7+4\sqrt{3}}-\sqrt{4+2\sqrt{3}}}$ is equal to ?

[CDS 2016(II)]

(a) 330 (b) 340
(c) 355 (d) 366

13. What is

$$\sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}}+\sqrt{1+\frac{1}{2^2}+\frac{1}{3^2}}+....+......\sqrt{1+\frac{1}{2007^2}+\frac{1}{2008^2}}$$

[CDS 2016(II)]

(a) $2008-\frac{1}{2008}$ (b) $2007-\frac{1}{2007}$
(c) $2007-\frac{1}{2008}$ (d) $2008-\frac{1}{2009}$

14. If x + $\frac{1}{1+\frac{1}{2+\frac{1}{3}}}$ =2, then what is x equal to?

[CDS 2016 (II)]

(a) $\frac{7}{10}$ (b) $\frac{13}{10}$
(c) $\frac{11}{10}$ (d) $\frac{17}{10}$

15. What is $\sqrt{\frac{0.064\times6.25}{0.081\times4.84}}$ equal to?

[CDS 2017(I)]

(a) 10/99 (b) 100/99
(c) 9 (d) 99

16. What is the value of $\sqrt[3]{4\frac{12}{25}}$ is?

[CDS 2017(II)]

(a) $1\frac{3}{5}$ (b) $1\frac{2}{5}$
(c) $1\frac{4}{5}$ (d) $2\frac{2}{5}$

17. Consider the following numbers :
1. 2222
2. 11664
3. 343343
4. 220347

Which of the above are not perfect squares?

[CDS 2017(II)]

(a) 1, 2 and 3 (b) 1, 2 and 4
(c) 2, 3 and 4 (d) 1, 3 and 4

18. The smallest integer with 4 digits which is a perfect square is

[CDS 2018(I)]

(a) 1000
(b) 1024
(c) 1089
(d) None of the above

19. What is the value of $\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}-\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$?

[CDS 2018(I)]

(a) $-2\sqrt{15}$ (b) $2\sqrt{15}$
(c) $\sqrt{15}$ (d) $-\sqrt{15}$

20. What is the value of $2+\sqrt{2+\sqrt{2+\sqrt{2+.....}}}$

[CDS 2019(I)]

(a) 1 (b) 2
(c) 3 (d) 4

21. What is the square root of $16 + 6\sqrt{7}$?

[CDS 2019(II)]

(a) $4 +\sqrt{7}$ (b) $4-\sqrt{7}$
(c) $3 + \sqrt{7}$ (d) $3-\sqrt{7}$

22. If $a=\sqrt{7+4\sqrt{3}}$, then what is the value of $a+\frac{1}{a}$

[CDS 2019(II)]

(a) 2 (b) 3
(c) 4 (d) 7

23. What is $\frac{1}{a^{m-n}-1}+\frac{1}{a^{n-m}-1}$?

[CDS 2020(I)]

(a) 1 (b) −1
(c) 0 (d) $2a^{m-n}$

24. If x = $\sqrt{2}$, y = $\sqrt[3]{3}$ and z = $\sqrt[6]{6}$ then which one the following is correct ?

[CDS 2020(I)]

(a) y < x < z (b) z < x < y
(c) z < y < x (d) x < y < z

25. What is the value of $\frac{1}{1+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+........+\frac{1}{\sqrt{99}+\sqrt{100}}$

[CDS 2020(II)]

(1) 1 (2) 5
(3) 9 (4) 10

26. If $x^m=\sqrt[14]{x\sqrt{x\sqrt{x}}}$, then what is the value of m?

[CDS 2020(II)]

(a) 1/8 (b) 1/4
(c) 3/4 (d) 7/4

27. $\frac{1}{1\times 2}+\frac{1}{2\times 3}+\frac{1}{3\times 4}+.........\frac{1}{n(n+1)}=\frac{99}{100}$ then what is the value of n ?

[CDS 2021(I)]

(a) 98 (b) 99
(c) 100 (d) 101

28. For what integral value of x is $\cfrac{12}{7-\cfrac{6}{7-\cfrac{3}{5-x}}}=x$?

[CDS 2021(I)]

(a) 4 (b) 3
(c) 2 (d) 1

29. What is the square root of $15-4\sqrt{14}$?

[CDS 2021(I)]

(a) $2\sqrt{2} -\sqrt{7}$ (b) $3\sqrt{2} -2\sqrt{17}$
(c) $\sqrt{15} -\sqrt{7}$ (d) $\sqrt{5} -\sqrt{3}$

30. What is the square root of $23-4\sqrt{15}$?

[CDS 2021(II)]

(a) $\sqrt{6} - 3\sqrt{2}$ (b) $7-3\sqrt{5}$
(c) $\sqrt{3} - 2\sqrt{5}$ (d) $\sqrt{5} - 4\sqrt{3}$

31. What is $\frac{1}{1+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{4}}+ \ldots + \frac{1}{\sqrt{2020}+\sqrt{2021}}$ equal to?

[CDS 2022(I)]

(a) $\sqrt{2020}+1$ (b) $\sqrt{2021}+1$
(c) $\sqrt{2020}+\sqrt{2021}-1$ (d) $\sqrt{2021}-1$

32. What is the smallest natural number from the following which must be subtracted from 9410 to make the remaining number a perfect square?

[CDS 2022(I)]

(a) 4 (b) 3
(c) 2 (d) 1

33. What is the value of the following?

$$\frac{1}{5\sqrt{4}+4\sqrt{5}}+\frac{1}{6\sqrt{5}+5\sqrt{6}}+\frac{1}{7\sqrt{6}+6\sqrt{7}}+\frac{1}{8\sqrt{7}+7\sqrt{8}}+\frac{1}{9\sqrt{8}+8\sqrt{9}}$$

[CDS 2022(I)]

(a) $\frac{1}{\sqrt{6}}$ (b) $\frac{1}{2}$
(c) 1 (d) $\frac{1}{6}$

34. If $x^2 - 20 = \sqrt{20+\sqrt{20+\sqrt{20+\sqrt{20+\text{....infinite terms}}}}}$, then what is x equal to?

[CDS 2022 (II)]

(a) 4 (b) 5

(c) $\sqrt{5}$ (d) $2\sqrt{5}$

35. If $x = \sqrt{4\sqrt{4\sqrt{4\sqrt{4.....}}}}$, then what is the value of x?

[CDS – 2023 (1)]

(a) 2 (b) 4
(c) 8 (d) 16

36. If the positive square root of $(5 + 3\sqrt{2})(5 - 3\sqrt{2})$ is α, then what is the positive square root of $8 + 2\alpha$?

[CDS – 2023 (1)]

(a) $2 + \sqrt{3}$ (b) $3 - \sqrt{2}$
(c) $\sqrt{7} - 1$ (d) $\sqrt{7} + 1$

37.

What is $\frac{3}{1^2 \times 2^2} + \frac{5}{2^2 \times 3^2} + \frac{7}{3^2 + 4^2} + ...$ equal to?

[CDS – 2024 (1)]

(a) 1 (b) 4
(c) 7 (d) 9

38.

If $\dfrac{1}{1+\dfrac{1}{b+\dfrac{1}{d+\dfrac{1}{e}}}} = \dfrac{421}{972}$,

then what is the value of a × b × c × d × e?

[CDS – 2024 (1)]

(a) 720 (b) 480
(c) 360 (d) 60

39.

If $x = \dfrac{1}{2+\dfrac{3}{4+\dfrac{5}{6+\dfrac{7}{8+\dfrac{9}{10}}}}}$

then which one of the following is correct ?

[CDS – 2024 (1)]

(a) $0 < x < 0.5$ (b) $x = 0.5$
(c) $0.5 < x < 1.0$ (d) $x > 1.0$

ANSWER KEY																			
1.	(b)	**2.**	(d)	**3.**	(a)	**4.**	(d)	**5.**	(c)	**6.**	(d)	**7.**	(d)	**8.**	(b)	**9.**	(a)	**10.**	(b)
11.	(b)	**12.**	(a)	**13.**	(a)	**14.**	(b)	**15.**	(b)	**16.**	(a)	**17.**	(d)	**18.**	(b)	**19.**	(a)	**20.**	(b)
21.	(c)	**22.**	(c)	**23.**	(b)	**24.**	(b)	**25.**	(c)	**26.**	(a)	**27.**	(b)	**28.**	(c)	**29.**	(a)	**30.**	(c)
31.	(d)	**32.**	(d)	**33.**	(d)	**34.**	(b)	**35.**	(b)	**36.**	(d)	**37.**	(a)	**38.**	(a)	**39.**	(a)		

CAPF PYQ

1. Which one of the following is the smallest number by which 2880 must be divided in order to make it a perfect square?

[CAPF 2017]

(a) 3 (b) 4
(c) 5 (d) 6

2. The least integer when multiplied by 2940 becomes a perfect square is:

[CAPF 2019]

(a) 10 (b) 15
(c) 20 (d) 35

ANSWER KEY

1.	c	**2.**	b																